LINUX / SYSTEMS

Kernel Basics Visual Guide

Syscalls, kernel vs userspace, modules, dmesg, and sysctl tuning.

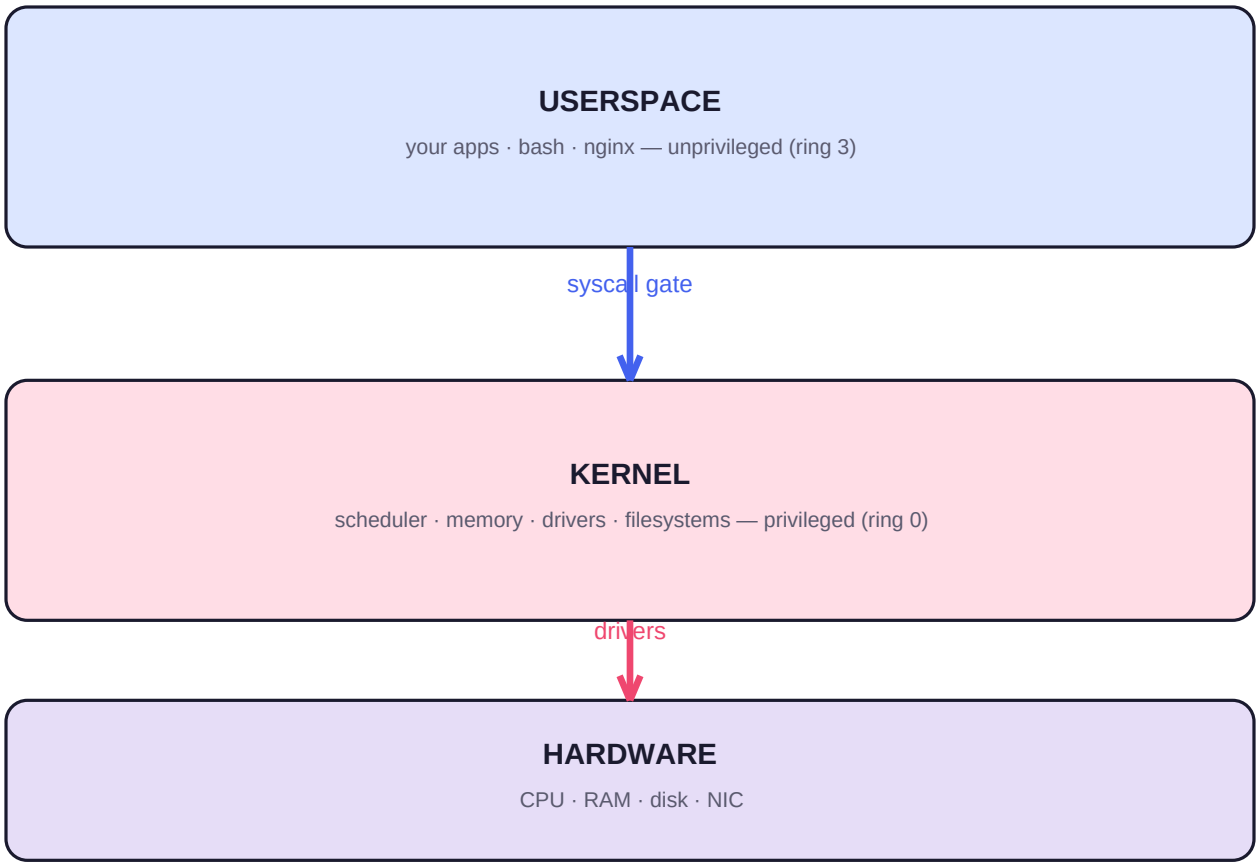
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Two Worlds: Userspace vs Kernel

Hardware is behind a wall; syscalls are the gate.



A syscall is a controlled trap into the kernel: validated arguments, privileged work, safe return. ~450 syscalls on x86_64 Linux — everything an app does with the outside world goes through them.

STEP 2/6

Syscalls in Action: strace

Watch a program talk to the kernel live.

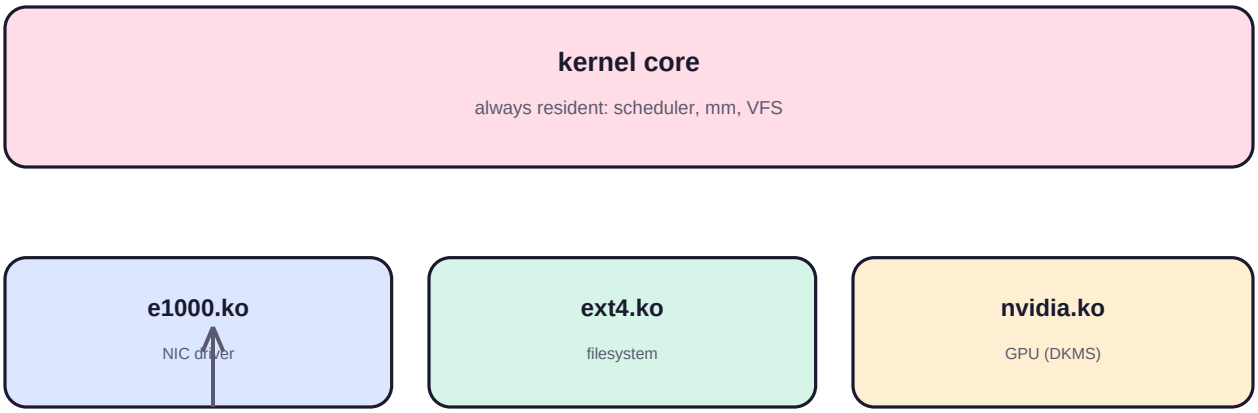
```
$ strace -c ls /tmp          # count syscalls, not just list
$ strace -e openat,read,write cat /etc/hosts
openat(AT_FDCWD, "/etc/hosts", O_RDONLY) = 3
read(3, "127.0.0.1 localhost\n", 131072) = 20
write(1, "127.0.0.1 localhost\n", 20) = 20
```

Syscall	Does
open/openat	open a file → file descriptor
read / write	move bytes via fd
fork / clone	create process/thread
execve	replace image with a program
mmap	map memory or files
socket/connect	networking

STEP 3/6

Kernel Modules: Drivers on Demand

The kernel is modular — drivers load and unload at runtime.




```
$ lsmod | head                # loaded modules
$ sudo modprobe e1000        # load one
$ sudo modprobe -r e1000     # unload
$ modinfo e1000 | grep -i parm   # its tunables
```

DKMS rebuilds out-of-tree modules (nvidia, zfs) automatically on kernel updates — without it, a kernel upgrade can leave you driverless at boot.

STEP 4/6

dmesg: The Kernel's Diary

Boot messages, driver probes, hardware errors — ring buffer.

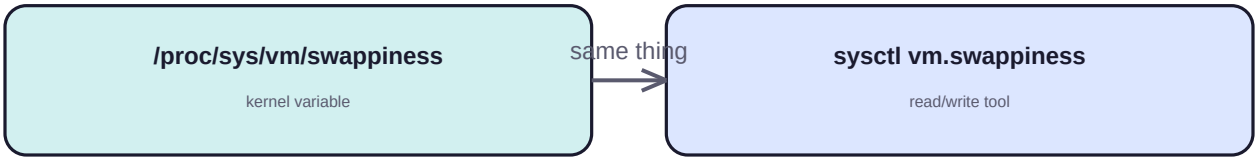
```
$ dmesg | head -20            # boot, from the start
$ dmesg -T | tail -30         # human timestamps, recent
$ dmesg -l err,warn          # only errors and warnings
$ dmesg -w                   # follow live (like tail -f)
```

- First suspect for 'new hardware not working': the driver probe lines are here.
- OOM killer, segfaults, thermal throttling, disk errors — all announced in dmesg.
- Ring buffer is finite: persistent logs live in the journal (journalctl -k).

STEP 5/6

sysctl: Tune the Kernel Live

/proc/sys is the control panel — sysctl is the safe handle.




```
$ sysctl vm.swappiness                      # read
$ sudo sysctl -w vm.swappiness=10           # write (until reboot)
$ sudo sysctl -w net.core.somaxconn=4096    # bigger listen backlog
# persist: /etc/sysctl.d/99-tuning.conf
vm.swappiness = 10
net.core.somaxconn = 4096
```

sysctl -w is instant but ephemeral; files in /etc/sysctl.d/ survive reboot. Change ONE knob, measure, then keep or revert — tuning by folklore breaks systems.

STEP 6/6

Kernel Versions & Upgrades

Know what you're running before you tune or debug.

```
$ uname -r              # 6.8.0-52-generic
$ hostnamectl | grep Kernel
$ ls /boot/vmlinuz-*   # installed kernels
```

- Versioning: major.minor-patch — distro kernels add their own suffixes and backports.
- Keep at least one older kernel installed: if the new one won't boot, GRUB → Advanced → old kernel.
- LTS kernels for servers (stability); mainline for new hardware support.
- Reboot into the new kernel to actually RUN it — installing without rebooting changes nothing.

STEP ✓/✓

Key Takeaways

Kernel Basics

- Userspace is unprivileged; every hardware/OS action crosses the syscall gate.
- strace reveals the syscall conversation — the ultimate 'what is it actually doing'.
- Modules load drivers on demand; DKMS keeps out-of-tree drivers alive across upgrades.
- dmesg is the kernel's diary: boot, drivers, OOM, hardware errors.
- sysctl tunes /proc/sys live; persist in /etc/sysctl.d and change one knob at a time.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev